

Lab 3: Hispanic Population Distribution in the Contiguous U.S. (2010 Census)

Shahreare Joy

January 31, 2025

Introduction:

Every 10 years, the US does a census where they count all the people living in the US. It provides information about things like population size, race, ethnicity, location, and other characteristics. This information can be used in many ways. For example, if a politician is getting votes from, let's say, the Asian population, they would see the census data and target certain areas where there are more Asian (attribute field DP0080006). If one wanted to know which state has the lowest number of African American population, they would be able to from this data; they would find that Montana has the lowest African American population. For my report, I will be using the information from the 2010 census to analyze the spatial distribution of the Hispanic population. I will only include the 48 US states and Washington, DC.

Data and Methods:

I started by obtaining the 2010 US census data from the US Census Bureau¹ website as a zip file. I then unzipped and imported the shapefile data into ArcGIS Pro. After, I converted the shapefile to NAD 1983 Contiguous USA Albers. I opened the Attributes table and removed Alaska, Hawaii, and Puerto Rico. It then had 49 rows, which included Washington, DC. I exported the features into new files. After, I used the data to generate four maps to show the distribution of the Hispanic population each with different visual and properties. For all, I used the primary symbology tab in the symbology pane.

The dot density map, figure 1, show a blue dot for every 30,000 Hispanic (attribute field DP0100002) of any origin in each of the states. The state abbreviations were shown and configured to show dark purple and yellow highlights to allow the user to see well. The color of the state borders was changed to black to allow for more differentiation. Each of the states has the same color, it is named yucca yellow.

A graduated symbol map was created, figure 2, using the same method as before but the type was changed to graduated symbol in the primary symbology tab. The field was set to Hispanic of any origin (attribute field DP0100002). It was then normalized using land area in square meters. The method I used was natural breaks and separated into 6 classes. Each of the classes was later recolored to allow for more differentiation between them other than just the size of the dot.

For my third map, I generated a red choropleth map or graduated color map. I used attribute field DP0100002 again but changed the normalization to the total population of each of the states (attribute field DP0010001). I then used the quantile method and separated it into 6 classes. I then changed the decimals into percentage using the advanced symbology option then format labels.

¹ United States Census Bureau. "State_2010Census_DP1.zip." July 19, 2012. Accessed January 28, 2025. https://www2.census.gov/geo/tiger/TIGER2010DP1/State_2010Census_DP1.zip.

For my final map, figure 4, I made a pie chart map to show the composition of the Hispanic population in each of the state by their origin. Using the Chart option from the primary symbology, I selected the four Hispanic subgroup. They include Mexican (attribute DP0100003), Puerto Rican, Cuban, and Other Hispanic or Latino. Their attribute increases by 1, respectively after the attribute number for Mexican. The map background color is the same as others, but the pie shown on the state is in four different dark colors. The color codes are shown by the legends on the left of the map to help the reader better understand what the map is showing.

Results:

My finding shows that there is an uneven distribution of the Hispanic population across the US states. The dot distribution map, figure 1, and percentage of Hispanics in the Contiguous US states, figure 3, show this finding. They were more concentrated around the more agricultural areas and the coasts. Of these states the top five most populated were California (14,013,719), Texas (9,460,921), Florida (4,223,806), New York (3,416,922), and Illinois (2,027,578). On the other hand, the top least populated were Vermont (9,208), North Dakota (13,467), Maine (16,935), South Dakota (22,119), and West Virginia (22,268). Figure 3 shows the Hispanic population in terms of the percentage of the total Hispanic population that lives in that state by area. You will find that DC has the highest density since it is a small area with larger numbers of Hispanics living there, which is recorded to be 54,749 Hispanic people.

The Hispanic population proportion by group in each of the U.S. state is shown by figure 4. This breaks the Hispanic group into four subgroups in pie graph format. Which includes Mexican, Puerto Rican, Cuban, and Other Hispanic or Latino. The most diverse group of Hispanic is seen on the East coast side of the US. Just to name a few, it includes states like Florida, New York, Pennsylvania, and Ohio. The central and west coast of America is majorly Mexicans. It includes states like California, Texas, Arizona, and more. Overall, figure 4 shows the diversity of different subgroups in the Hispanic population.

From the attribute table, I found that the least populated state was Wyoming with a population size of 563,626 people. I wanted to check to see which states had Hispanic population greater than the least populated state, in this case it was Wyoming. To do this, I opened the attribute table and clicked selecting by attribute where the Hispanic population was more than of Wyoming. The states that have a Hispanic population more than this least populated state are NM, NV, CO, AZ, MA, WA, VA, NJ, NC, GA, PA, IL, FL, NY, TX, and CA.

Discussion:

In figure 1, the dots distribution shows that the dots are more concentrated in states like CA, TX, IL, NY, NJ, and FL. This could be because of the agricultural work they do. States like California and Illinois are also known for their farmland and agriculture. They probably work on the farm doing physical labor in picking things. Texas, New Jersey, and Florida are similar cases.

From this, my understanding is many of them could be undocumented and or does not know how to speak English and so this type of labor gives them the wealth they need to survive and feed

their family. If more data for as to what type of work, they do would have helped me in making a more precise conclusion as to why they are more populated in these types of states. Additionally, getting the population analysis from a county level could make it clearer as to which areas they live in, in different states and what those counties have in common. This would help in deciding as to why they are concentrated in the places they are at.

Conclusion:

The United States of America has one of the most diverse groups of people in its population poll. Each of the group lives in different parts of the country; some are more concentrated in one area; well others are not. For this lab, I looked at the US population of Hispanics at a state level. Each of the states had a distinct number of Hispanics that lived there during the 2010 U.S. State census. From the data, it was not clear as to what area of the state they are more concentrated in, but it could be seen by maps like figure 1 as to how many of them and in which states they were more concentrated in. It could be further distinguished by maps like figure 4, to show the sub-group of each of the Hispanic populations and their majority in the state from that total Hispanics of that state. Overall, the Hispanic population is everywhere but they are more concentrated in places that are more into farming. If more specific data from the county level were available, it could be further distinguished as to why they live there.

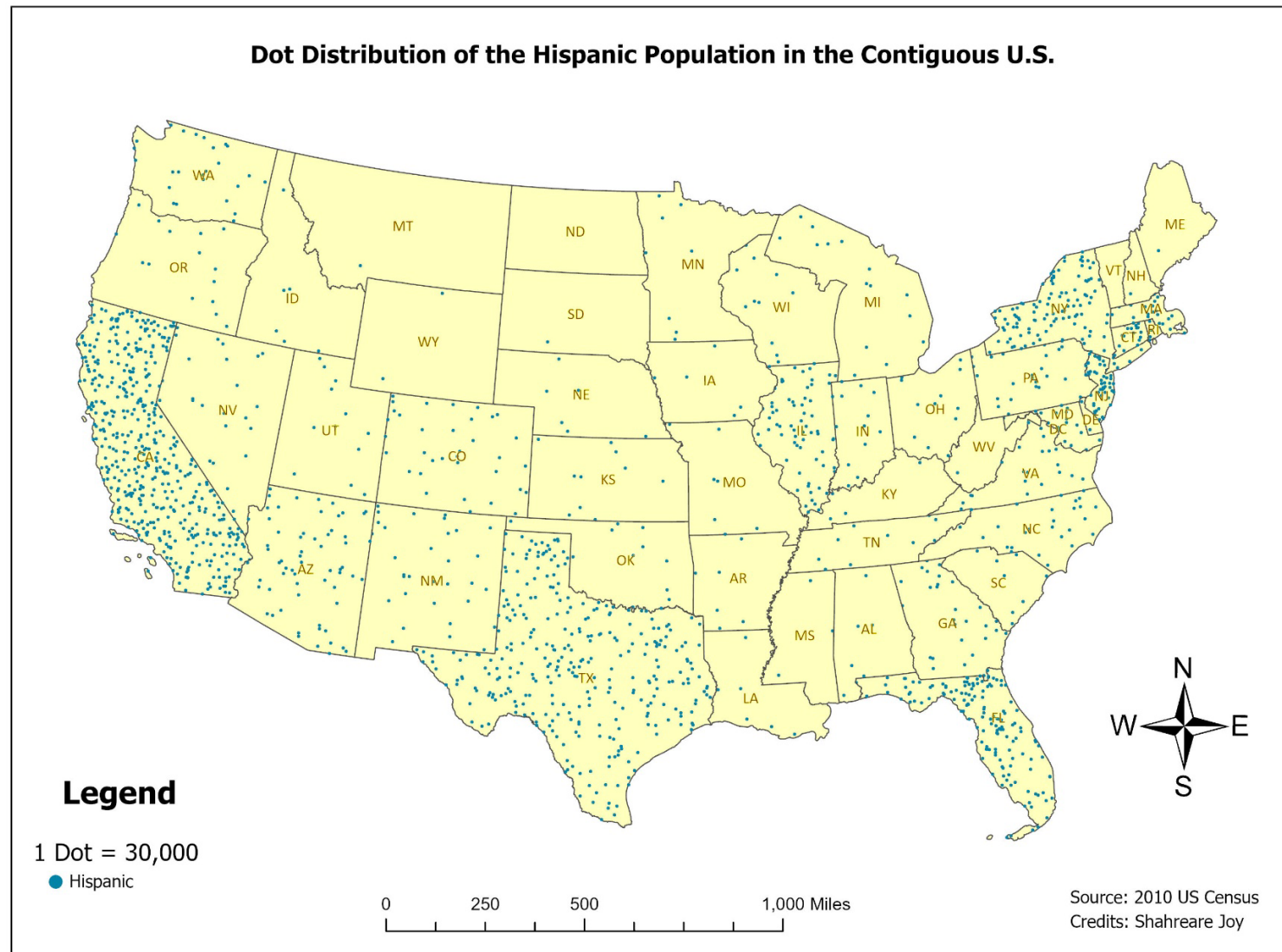


Figure 1: A dot distribution of the Hispanic population in the contiguous U.S. where each dot representing 30,000 Hispanics.

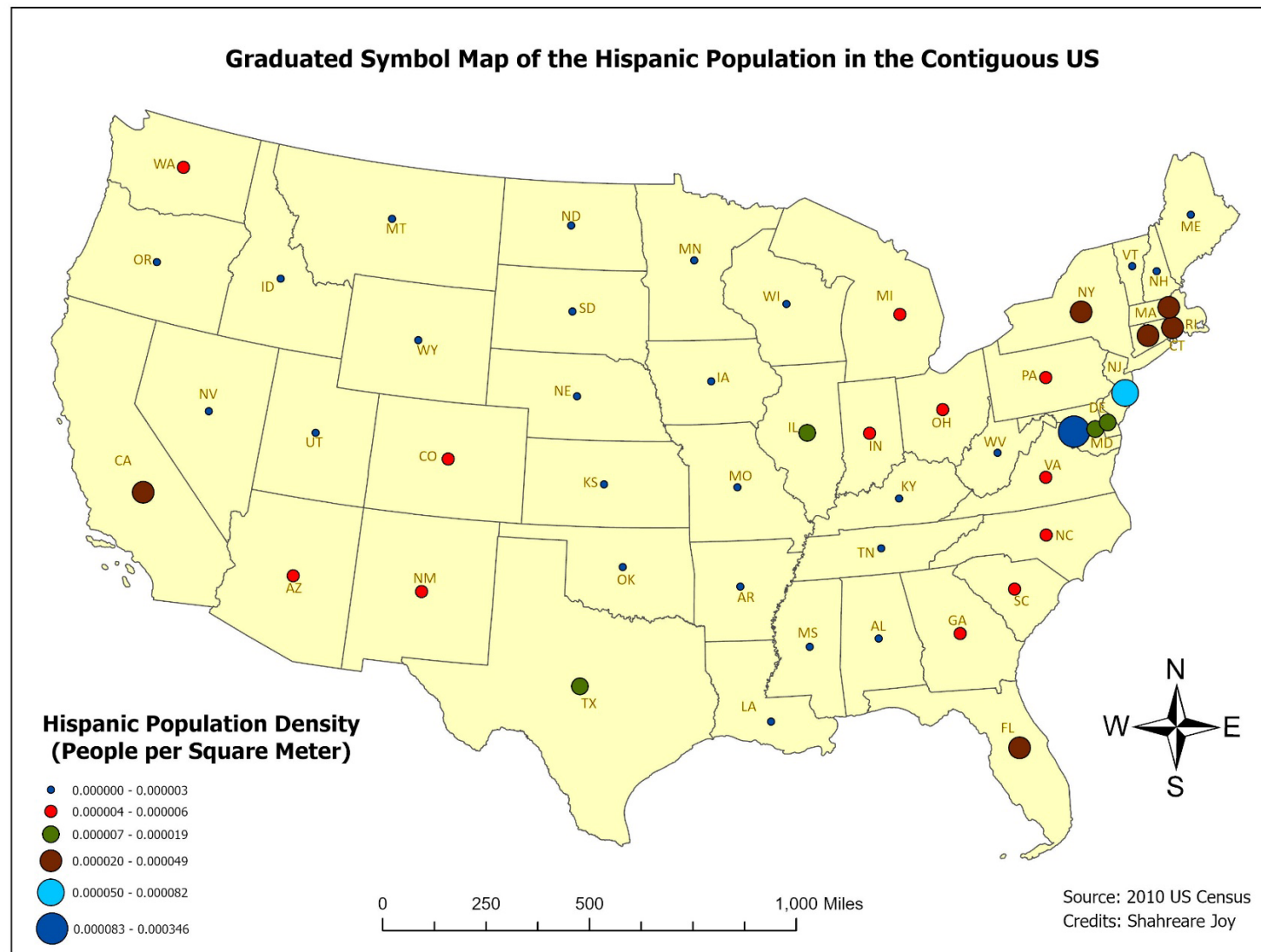


Figure 2: A graduated symbol map showing the Hispanic population in each state according to its land size in square meters.

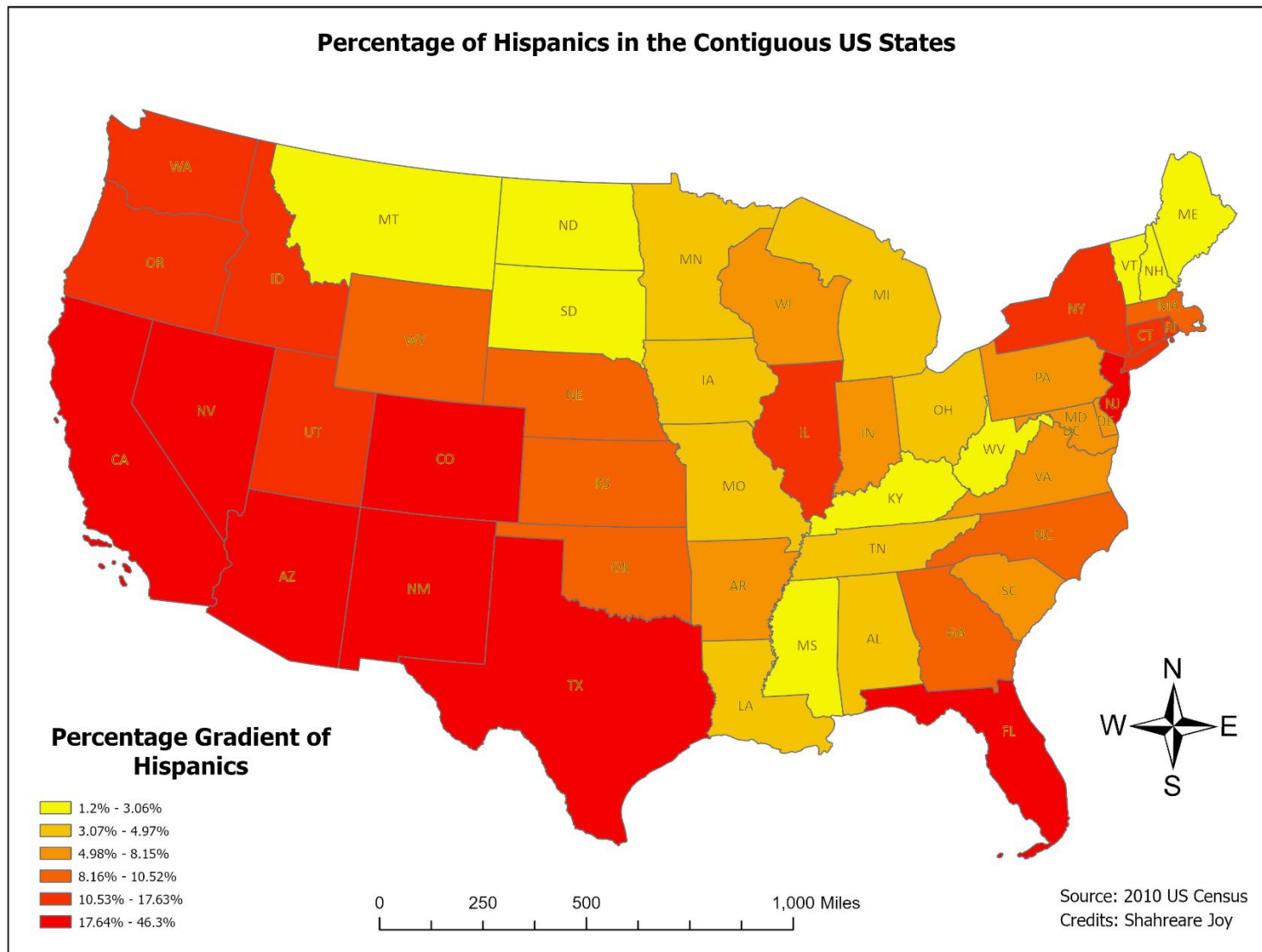


Figure 3: A percent gradient map of the Hispanic population relative to the total population of each state.

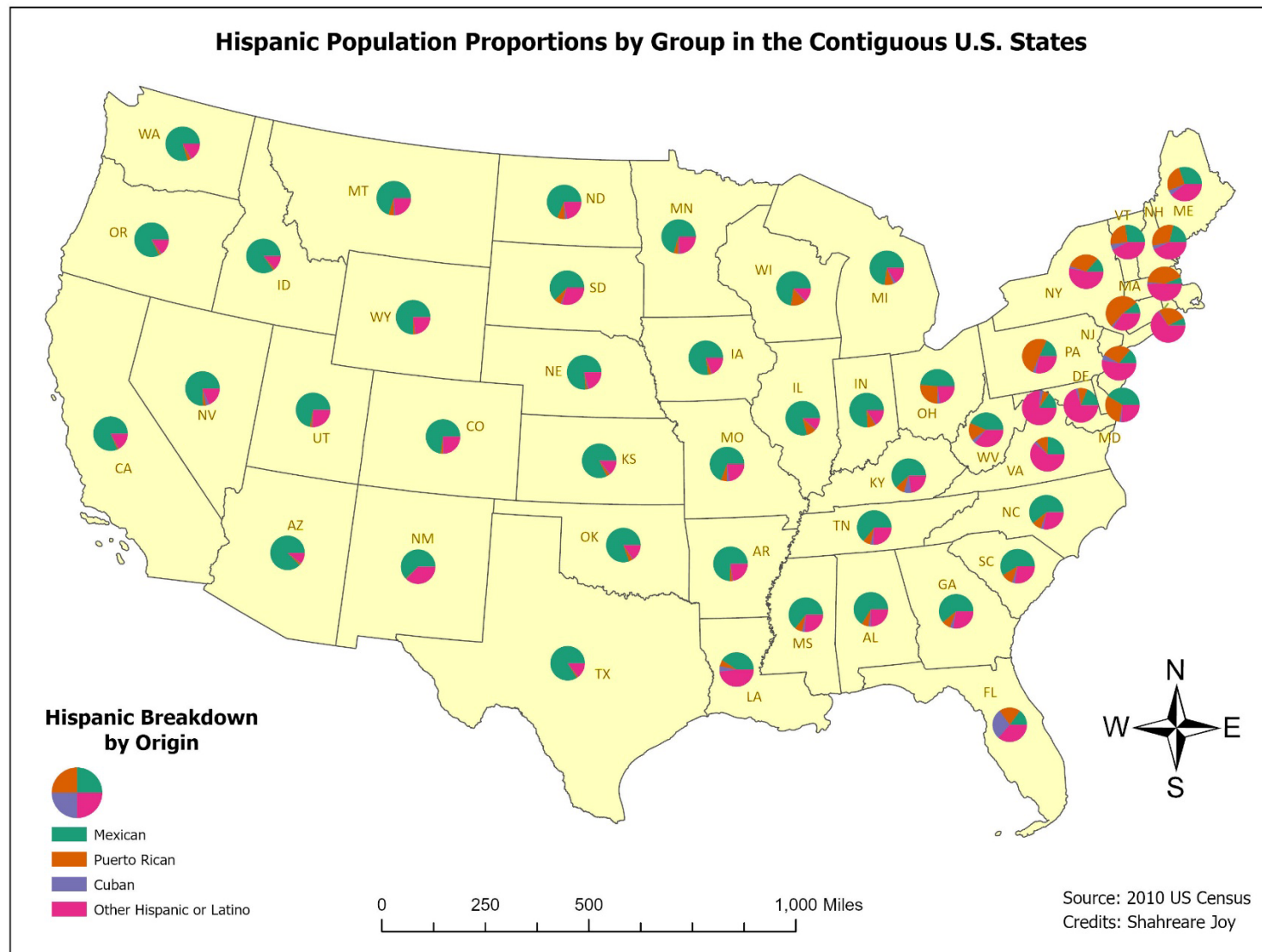


Figure 4: Pie chart graph of Hispanic population proportions by origin across the contiguous U.S. states.